

MX17 drift window & drift voltage

measurement-driven sizing of the DREAM readout for beam data

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2026-07-19 (run_54 cosmics + run_55 beam, nTOF EAR2)

TL;DR: latency 35 → **32**, n_samples 32 → **28**, det A drift 600 → **700 V** (staged), B/C/D stay 800 V, short on-beam det-A drift scan as the 700 V spark gate.

The question and the constraints

Decide for the 4-det Micromegas stack:

- ▶ minimum `n_samples` (60 ns DREAM sampling) with no drifting primary lost on *any* detector;
- ▶ drift HV: det A spark-limited (sparked @800 V, stable @600 V) — push to 700?

Hard facts

- ▶ 30 mm gap, Ar/iC₄H₁₀ 90/10, all four dets;
- ▶ **series gas line A→B→C→D** (wetness rises down-chain);
- ▶ current: $32 \text{ smp} \times 60 \text{ ns} = 1.92 \text{ } \mu\text{s}$, latency 35;
- ▶ drift: A 600 V, B/C/D 800 V; resist 560 V.

Data used (2026-07-18)

- ▶ run_54: cosmics, 4 subruns, 9k evts/FEU;
- ▶ **run_55: beam** (19:11, Mode-3 scint-doubles, r560 subruns) — real trigger, real flash;
- ▶ Magboltz: clean 90/10 @CERN P + H₂O grid.

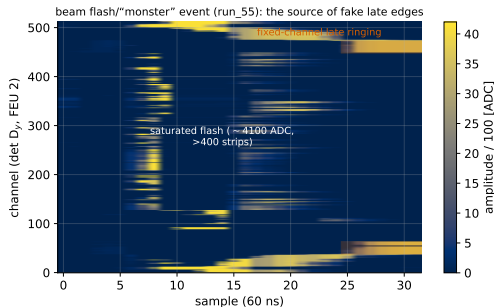
June lesson: wet gas ($\sim 1\%$ H₂O) *halves* low-field drift speed; the 500 V June point truncated its window.

Method: de-noising is the whole game

Extractor (24_drift_time_edges.py)

- ▶ common mode per 64-ch DREAM chip (noise is per-connector), 2 iterations, track strips masked;
- ▶ per-strip robust MAD threshold + width ≥ 2 , hot-channel mask;
- ▶ spatial clusters (≥ 3 adjacent strips) = track;
- ▶ per event: prompt (earliest peak), deep edge (latest peak), pulse rise/end, span.

The decisive cut: events with >100 hit strips (“monster” /flash, $\sim 2\%$ of beam triggers; det-D sparks in cosmoics) are excluded.



Finding 1 — the “slow/late B & D” of the first look was artifact

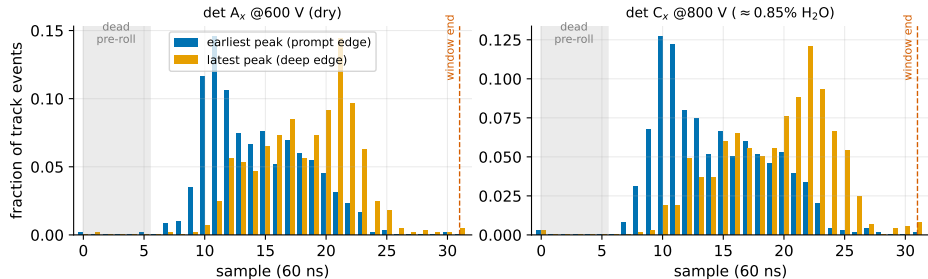
- ▶ Saturated flash events are followed by **fixed channel blocks** (e.g. FEU2 ch 38–63, 448–472) ringing at samples 24–31 $\Rightarrow$ they masqueraded as 25-strip “late clusters”. Every inspected late cluster was one of these.
- ▶ det D cosmics: 129–152 spark events / 9000 $\Rightarrow$ after the cut, D_x ceiling fraction 8.9% $\rightarrow$ 1.5%.
- ▶ det B looks contaminated only in the cosmic run (coherent noise); its *beam* x-plane matches C/D. No evidence B is slow.

ceiling fraction (peak $\geq$ smp 31)	A_x	C_x	D_x	B_x
raw first look (handoff)	0.3%	0.6%	7–20%	21–30%
de-noised + monster cut (cosmics)	0.3%	0.4%	1.5%	(noise-limited)

No detector is losing drifting primaries at 32 smp / latency 35 today.

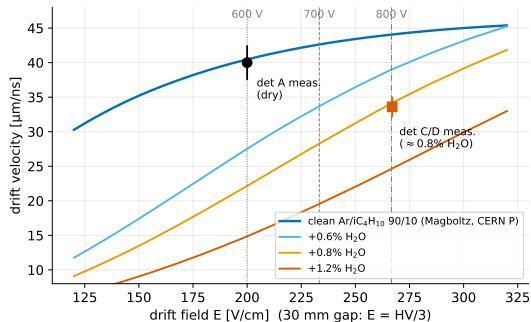
Finding 2 — timing is universal: prompt at sample 8–9, both triggers

run_54 cosmics, de-noised track clusters — latency 35



- First signal above threshold: sample 6–7 (p5); prompt *peak*: 8–9 $\Rightarrow$ peak $\approx$ latency–26.
- Identical in cosmic-singles and beam Mode-3 triggers, all four detectors — the suspected per-detector t_0 offsets were cluster-geometry artifacts. **One latency serves both run types.**
- $\Rightarrow$ ~ 6 dead pre-roll samples reclaimable, but *only* by lowering latency.

Finding 3 — water: A is dry; B, C, D all breathe $\sim 0.8\%$ H_2O

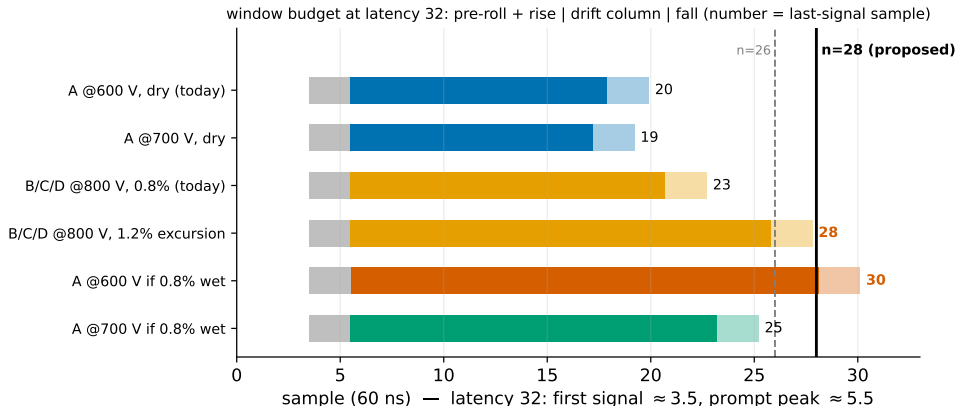


Span of ≥ 6 -strip clusters (cosmics, p95),
recorded-column fraction 0.89 anchored on
det A = clean:

det	HV [V]	T_{full} [smp]	v [$\mu\text{m/ns}$]	H_2O
A	600	12.4	≈ 40	$\leq 0.2\%$
C	800	15.2	33.0	0.85%
D	800	14.6	34.2	0.80%
B	800	~ 15	~ 33	$\sim 0.8\%$

- Chain wetting **saturates at the first hop**: no $A \rightarrow D$ slope, $B \approx C \approx D$ near June's $\sim 1\%$ equilibrium.
- Deep-charge attenuation mild: $\lambda_{\text{eff}} \approx 23\text{--}25$ mm (June: 16–18).

Window budget $\Rightarrow$ latency 32, n_samples 28



- **n=28 @ latency 32:** covers today's worst deep edge with ~ 4 smp margin, survives a wetness excursion to $\sim 1.1\%$; saves 12.5% readout/size. Same config works for cosmics.
- n=26: OK for today's gas, zero excursion headroom — only with per-run span monitoring.
- **16–20 samples impossible:** the B/C/D drift column alone is ~ 15 smp at 0.8% H_2O .
- Dry the gas line $\rightarrow$ all columns ~ 12 smp $\rightarrow$ n=22–24 would become safe. **A dryer is worth more than any further trim.**

Det A: 600 $\rightarrow$ 700 V (staged) — insurance, not speed

Why

- ▶ Today A is dry: 700 V buys only 0.7 smp. Irrelevant.
- ▶ **The one catastrophic corner:** A@600 V in gas as wet as B/C/D *already are* (0.8%) $\Rightarrow$ 22.6 smp column $\Rightarrow$ truncates any affordable window (June's 500 V failure mode).
- ▶ A@700 V in the same gas: 17.7 smp $\Rightarrow$ fits $n=28$.
- ▶ A is first in the chain, but one flow drop / line disturbance makes it wet overnight.
- ▶ Physics cost $\approx$ nil (micro-TPC z-sampling -6%).

How (spark risk: sparked @800 V)

1. one subrun at **650 V**, watch `hv_monitor.csv` (trips, $I > 0.2 \mu\text{A}$);
2. then **700 V** soak under beam;
3. fallback 650 V (still halves the wet-corner risk).

B/C/D: keep 800 V — wetter gas needs the field; nothing gained by lowering.

Proposed on-beam drift scan (small, cheap, do it)

- ▶ **det A only:** 600 V (reference exists) $\rightarrow$ 650 $\rightarrow$ 700, ≥ 30 min each during normal Mode-3 running. B/C/D untouched $\Rightarrow$ their physics data continues.
- ▶ Yields per point: ~ 100 – 200 clean A-tracks/plane $\rightarrow$ span to ± 1 smp.
- ▶ Deliverables:
 1. **spark/current stability at 650/700 = the actual gate** for the HV move;
 2. span(600)/span(700) ratio pins A's water content assumption-free (dry: 1.06, 0.8%-wet: 1.28 — cleanly separable);
 3. confirms the predicted $\sim$ sample-20 endpoint before freezing latency 32 / n=28.
- ▶ Optional, low priority: one B/C/D point at 750 V (wet-gas $v(E)$ slope); costs their margin while running.

Summary & open items

knob	recommendation
latency	35 → 32 (prompt peak → smp 5–6)
n_samples	32 → 28 (floor 26 with monitoring; <24 needs drier gas)
det A drift	700 V, staged 650 → 700 with HV-monitor soak
B/C/D drift	keep 800 V
beam drift scan	yes — det A 600/650/700, 30 min each

- ▶ Open (non-blocking): v6 prompt-anchored verification pass + run_53 16-h gas-stability cross-check — computed, JSONs on `daq_lxplus:/tmp/v6_*.json` (box was off-network); expected to tighten y-plane numbers only.
- ▶ Electronics follow-up: post-saturation ringing blocks (fixed per-FEU channel ranges); window trim already cuts most of it offline.
- ▶ Full writeup: `mx_july_beam_qa/DRIFT_WINDOW_ANALYSIS.md`.